

OpenShift 4.20 — IPI Cluster Build on vSphere (Airgapped, Default/Internal LB, Static IPs, No External LB)

Assumes: mirror registry VM already built (per your existing procedure), single vCenter / single datacenter / single cluster, no DHCP, static IPs reserved, 3 control-plane + 2 compute, **no external load balancer** — using the installer's default managed load balancer (`OpenShiftManagedDefault`), which runs keepalived + haproxy *inside* the cluster nodes themselves.

1. GUI vs CLI — set expectations first

There is no full point-and-click GUI installer for vSphere IPI in a disconnected environment — the actual `openshift-install create cluster` step is always a CLI command, run from your bastion. This is true regardless of vendor tooling; it isn't something being withheld from you.

What genuinely is GUI-driven:

- All vCenter-side prep (folder/resource pool creation, permissions, networking, OVA/Content Library handling) — done entirely in the vSphere Client GUI.
- DNS record creation — GUI in whatever DNS server product you use.
- Cluster verification and day-2 operations after install — the OpenShift web console is a full GUI from that point forward.

The one alternative with an actual browser-based install wizard is the **Assisted Installer** (`console.redhat.com` hybrid cloud console, or a locally-hosted instance for disconnected use). It's a different install mechanism (ISO-boot based, not `openshift-install` -driven IPI), has its own vCenter integration, and is a valid option if a GUI wizard genuinely matters more to you than the IPI method itself — happy to write that up separately if you want to switch tracks. Everything below assumes you're staying with IPI as decided.

2. Content Library — can you pre-upload the OVA? Still no,

same reason as before

Worth restating since this doc stands alone: **the IPI installer uploads and imports the RHCOS OVA itself, every time, from the `clusterOSImage` URL in `install-config`**. It does not consume a pre-staged Content Library item for that primary import — this applies with or without an external LB, it's unrelated to your LB choice.

If you still want the OVA available in a Content Library for other purposes (manual VM builds, reference, compliance requirement to have images cataloged), here's the GUI procedure — just know the installer won't use it:

1. In the vSphere Client, go to **Menu → Content Libraries**.
2. Click **Create** (if you don't have one already) → name it (e.g. `rhcos-images`) → choose a datastore → Finish.
3. Open the library → **Actions → Import Item**.
4. Choose **Local file** → browse to your downloaded `rhcos-4.20.0-x86_64-vmware.x86_64.ova` → Import.
5. Once imported, you can right-click the item → **New VM from This Template** any time you need a manual RHCOS VM — but again, `openshift-install` will not reference this item.

For the install itself, host the OVA on an internal HTTP server instead (same as before):

```
mkdir -p /var/www/html/rhcos && cp rhcos-4.20.0-x86_64-vmware.x86_64.ova /var/www
# nginx/httpd, or for quick testing:
python3 -m http.server 8080 --directory /var/www/html
```

3. No external LB — what changes

With `loadBalancer.type: OpenShiftManagedDefault` (the default — you simply omit the `loadBalancer:` stanza entirely), the installer:

- Runs **keepalived** on the 3 control-plane nodes to own the **API VIP** via VRRP.
- Runs **keepalived** on the nodes hosting the default Ingress Controller (workers, by default, since you have 2) to own the **Ingress VIP**.
- Runs an internal **haproxy** on each control-plane node, load-balancing to all control-plane API servers.

You need two separate, free IP addresses for `apiVIPs` and `ingressVIPs` — unlike the external-LB case, these cannot be the same IP. Two independent keepalived VRRP instances trying to answer ARP for the same IP on the same L2 segment will conflict (flapping ownership, intermittent unreachability) — this is a hard technical constraint, not a preference. Reserve two IPs, neither assigned to any host, both inside your `machineNetwork` CIDR.

Also reserve a **VRRP router ID (vrid)** mentally — the installer auto-assigns one, but if you have *other* keepalived-based systems (F5, other OpenShift clusters, HA Linux services) on the same L2 segment/VLAN, VRID collisions cause the same ARP-flapping symptom. Worth checking with your network team if this VLAN hosts anything else running keepalived.

4. DNS entries required

Base domain example: `example.local`, cluster name `ocp1`. Same PTR requirement as static-IP installs generally — RHCOS needs reverse DNS to self-assign hostnames correctly at boot with no DHCP.

Hostname	Type	Points to / Value	Purpose
<code>api.ocp1.example.local</code>	A	API VIP (reserved, e.g. 10.10.10.10)	API server access — external + internal
<code>api-int.ocp1.example.local</code>	A	API VIP (same as above)	Internal-only API access used by nodes
<code>*.apps.ocp1.example.local</code>	A	Ingress VIP (reserved, e.g. 10.10.10.11)	Wildcard for all routes/applications
<code>bootstrap.ocp1.example.local</code>	A	bootstrap static IP	Bootstrap node resolution during install
<code>bootstrap.ocp1.example.local</code>	PTR	bootstrap static IP → hostname	Correct hostname assignment (no DHCP)
<code>master-0/1/2.ocp1.example.local</code>	A	control-plane static IPs	Control-plane node resolution
			Correct hostname

master- 0/1/2.ocpl.example.local	PTR	control-plane IPs → hostnames	assignment (no DHCP)
worker- 0/1.ocpl.example.local	A	compute static IPs	Compute node resolution
worker- 0/1.ocpl.example.local	PTR	compute IPs → hostnames	Correct hostname assignment (no DHCP)
bastion.example.local	A	bastion static IP	Installer/CLI host resolution
mirror- registry.example.local	A	mirror registry VM IP	Image pulls during and after install
vcenter.example.local	A	vCenter IP	Installer → vCenter API calls

No `etcd-*` SRV records needed for IPI (UPI-only requirement). No HAProxy/LB VM entry needed this time since there isn't one.

5. install-config.yaml (default/internal LB, static IPs, single zone/cluster)

```

apiVersion: v1
baseDomain: example.local
metadata:
  name: ocpl
platform:
  vsphere:
    apiVIPs:
      - 10.10.10.10
    ingressVIPs:
      - 10.10.10.11
  # loadBalancer stanza omitted entirely -> defaults to OpenShiftManagedDefault
clusterOSImage: http://mirror-registry.example.local:8080/rhcos/rhcos-4.20.0-
vCenters:
  - server: vcenter.example.local
    user: svc-ocp-installer@vsphere.local
    password: '<vcenter-password>'
    datacenter: DC1

```

```
failureDomains:
- name: dc1-cluster1
  region: region1
  zone: zone1
  server: vcenter.example.local
  topology:
    datacenter: DC1
    computeCluster: /DC1/host/Cluster1
    resourcePool: /DC1/host/Cluster1/Resources
    networks:
      - ocp1-network
    datastore: /DC1/datastore/ds1
    folder: /DC1/vm/ocp-420
hosts:
- role: bootstrap
  networkDevice:
    ipAddrs: ["10.10.10.5/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
- role: control-plane
  networkDevice:
    ipAddrs: ["10.10.10.20/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
- role: control-plane
  networkDevice:
    ipAddrs: ["10.10.10.21/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
- role: control-plane
  networkDevice:
    ipAddrs: ["10.10.10.22/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
- role: compute
  networkDevice:
    ipAddrs: ["10.10.10.30/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
- role: compute
  networkDevice:
    ipAddrs: ["10.10.10.31/24"]
    gateway: 10.10.10.1
    nameservers: ["10.10.10.2"]
controlPlane:
  name: master
  replicas: 3
```

```

platform:
  vsphere:
    cpus: 8
    coresPerSocket: 2
    memoryMB: 16384
    osDisk:
      diskSizeGB: 120
compute:
  - name: worker
    replicas: 2
    platform:
      vsphere:
        cpus: 4
        coresPerSocket: 2
        memoryMB: 16384
        osDisk:
          diskSizeGB: 120
networking:
  networkType: OVNKubernetes
  clusterNetwork:
    - cidr: 10.128.0.0/14
      hostPrefix: 23
  serviceNetwork:
    - 172.30.0.0/16
  machineNetwork:
    - cidr: 10.10.10.0/24
pullSecret: '<merged pull-secret.json, one line>'
sshKey: '<bastion public ssh key>'
additionalTrustBundle: |
  -----BEGIN CERTIFICATE-----
  <mirror registry CA cert>
  -----END CERTIFICATE-----
imageDigestSources:
  - source: quay.io/openshift-release-dev/ocp-release
    mirrors:
      - mirror-registry.example.local:8443/openshift/release-images
  - source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
    mirrors:
      - mirror-registry.example.local:8443/openshift/release
  - source: registry.redhat.io/redhat/redhat-operator-index
    mirrors:
      - mirror-registry.example.local:8443/redhat/redhat-operator-index

```

Both `apiVIPs` and `ingressVIPs` must sit inside `machineNetwork` (`10.10.10.0/24` above), free of any other assignment. Copy real `imageDigestSources` values from your `oc-mirror` output as before.

6. Cluster build procedure

```
# 1. Backup install-config before the installer consumes it
mkdir ocp1-install && cp install-config.yaml ocp1-install/
cp install-config.yaml install-config.yaml.bak

# 2. Kick off the install
openshift-install create cluster --dir=ocp1-install --log-level=info

# 3. Watch bootstrap in a second terminal
openshift-install wait-for bootstrap-complete --dir=ocp1-install --log-level=info
```

No manual LB backend edits needed here — keepalived on the bootstrap+masters answers the API VIP automatically as soon as the nodes are up; the installer removes bootstrap from rotation itself once control-plane is healthy.

```
# 4. Destroy the bootstrap VM once bootstrap-complete returns
openshift-install destroy bootstrap --dir=ocp1-install

# 5. Log in and approve worker CSRs
export KUBECONFIG=ocp1-install/auth/kubeconfig
oc get csr -o name | xargs oc adm certificate approve
# repeat once or twice — a second batch of CSRs appears per node after the first

# 6. Wait for full install completion
openshift-install wait-for install-complete --dir=ocp1-install --log-level=info

# 7. Wire in the mirrored operator catalog (from your oc-mirror output)
oc apply -f working-dir/cluster-resources/imageDigestMirrorSet.yaml
oc apply -f working-dir/cluster-resources/catalogSource-redhat-operator-index.yaml

cat <<EOF | oc apply -f -
apiVersion: config.openshift.io/v1
kind: OperatorHub
metadata:
  name: cluster
spec:
  disableAllDefaultSources: true
EOF
```

```
# 8. Verify
oc get co                                # all Available=True, Degraded=False, Progress
oc get nodes -o wide                     # 3 masters + 2 workers, Ready
oc get clusterversion
# confirm which node currently owns each VIP:
oc debug node/<a-control-plane-node> -- chroot /host ip a | grep 10.10.10.10
oc debug node/<a-worker-node> -- chroot /host ip a | grep 10.10.10.11
```

7. NSX-level ports required

Source	Destination	Port	Purpose
Bastion	vCenter	443/TCP	Installer → vCenter API for VM provisioning
Bastion	vCenter	902/TCP	NFC file transfer (OVA import into vCenter)
Bastion	All cluster nodes	22/TCP	SSH (troubleshooting, <code>core</code> user)
Bastion	API VIP	6443/TCP	<code>openshift-install</code> / <code>oc</code> → API
Bastion	Mirror registry	8443/TCP	Image pulls/pushes during mirror steps
Bastion	HTTP OVA host	8080/TCP	Only if hosted separately from bastion
All nodes	API VIP	6443/TCP	Nodes → API server
All nodes	API VIP	22623/TCP	Nodes → Machine Config Server
Control-plane nodes ↔ each other	—	112/VRRP (proto 112, not TCP/UDP)	keepalived VRRP for API VIP
Worker (ingress) nodes ↔ each other	—	112/VRRP	keepalived VRRP for Ingress VIP
All nodes ↔ all nodes	—	2379-2380/TCP	etcd client + peer traffic (masters only)

All nodes ↔ all nodes	—	6081/UDP	OVN-Kubernetes Geneve overlay
All nodes ↔ all nodes	—	9192/TCP	OVN-Kubernetes metrics
All nodes ↔ all nodes	—	9000-9999/TCP+UDP	Host-level node services
All nodes ↔ all nodes	—	10250-10259/TCP	Kubelet, scheduler, controller-manager
All nodes ↔ all nodes	—	30000-32767/TCP+UDP	NodePort service range
All cluster nodes	Mirror registry	8443/TCP	Pulling release + operator images
All cluster nodes	DNS server	53/TCP+UDP	Name resolution
All cluster nodes	NTP source	123/UDP	Time sync — cert validation depends on it
All cluster nodes	vCenter	443/TCP	vSphere CSI / cloud-provider API calls
Admin workstation	API VIP	6443/TCP	<code>oc /console</code> API access
Admin workstation	Ingress VIP	443/TCP	Web console + application access

Protocol 112 (VRRP) is easy to miss in NSX policies built around TCP/UDP service objects — make sure it's explicitly allowed, not just assumed as part of "all traffic between cluster nodes," since some NSX DFW rule templates only enumerate TCP/UDP.

8. Production-safety notes

- **Two free IPs, not one, and both inside `machineNetwork`** . Re-stated because it's the single most common mistake when moving from an external-LB design to the default LB.
- **NTP is not optional** — same as always; cert validation failures from clock drift look like random network problems.
- **etcd backups from day one** via the standard `cluster-backup.sh` process.

- **Rotate/secure the vCenter service-account credential** — sits in `install-config.yaml` and in the cluster's `vsphere-creds` secret post-install.
 - **The default LB has no cross-node redundancy beyond keepalived's own failover** — that's expected and fine (it's designed for exactly this), but understand that all control-plane nodes must stay on the same L2 segment for VRRP to work; if you ever plan multi-segment control plane, you'd need the external-LB design instead.
 - **Approve CSRs deliberately**, not via a permanent auto-approver, in a disconnected/security-sensitive environment.
 - **Confirm the static-IP `hosts:` field's exact support status on your specific 4.20.z build** — `openshift-install explain installconfig.platform.vsphere.hosts`, and check release notes for whether `featureSet: TechPreviewNoUpgrade` is still required.
 - **Don't skip `wait-for install-complete`** as your definition of "installed."
 - **Day-2 scaling with static IPs** needs Machine API + IPAddressClaim/IPAM integration — different mechanism than the day-1 `hosts: list`, plan separately before you need to scale.
-

9. Reference links

- [Installing on VMware vSphere — OCP 4.20 overview](#)
- [Installer-provisioned infrastructure chapter — OCP 4.20](#) — if this redirects, use the version selector on the page for 4.20
- [User-provisioned infrastructure chapter — OCP 4.20](#) (context/comparison only — you're on IPI)
- [Installing on vSphere using the Assisted Installer — OCP 4.20](#) (the GUI-wizard alternative mentioned in section 1)
- [OCP 4.20 release notes](#)
- [OCP 4.20 Installation overview](#)
- [Red Hat: How to perform an IPI installation of OpenShift on vSphere with static IP addresses](#) — background on the `hosts:` field and its Tech Preview origins
- [OKD 4.20 — Installing a cluster in a disconnected environment \(vSphere UPI, useful for mirroring concepts\)](#)
- [openshift/installer GitHub — external load balancer docs](#) — `loadBalancer` stanza

background (OpenStack-titled doc, but the stanza and
UserManaged / OpenShiftManagedDefault semantics are shared across platforms)